DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
UNIVERSITY COLLEGE OF ENGINEERING(A)
Osmania University, Hyderabad-07
CLASS TEST-I

SUBJECT : Data Analytics using R
Class: B.E (CSE)
Semester: IV

Max Marks: 30
Duration: 1.5 Hour

	PART A Answer All Questions (10 M)	Marks	BT	CO
1.a)	What are the advantages of R over other programming languages?	1	2	1
b)	Write the syntax of while loop and for loop in R? Give example for each.	2	2	1
c)	What is a list in R and write the syntax and examples?	2	1	1
d)	What is the output of the given R statements: **a = c(10, 20, 30, 40, 50, 60)** **n = array(a, dim = c(2, 2, 3))** **print(n)**	1	2	1
e)	Name any two plotting functions in R and give example for each.	1	3	1
f)	What is the output of the given R program? **m <- matrix(c(10,20,30,40,50,60), nrow=4,ncol=3,byrow=FALSE)** **print(m)** **cat("hello", m)**	2	1	2
g)	Write the output of the R program a <- c(11,12,34,42,50) b <- c("R", "Is", "Fun!","Let's","Learn") c <- c(TRUE,FALSE,TRUE,TRUE,FALSE) data.frame(a,b,c) my_frame <- data.frame(a,b,c) print(my_frame)	1	1	2
	PART B **Answer Any Two Questions (20 M)**			
2.	a) Explain different IDEs used in R programming and data types in R. b) Write a program to find the sum of first 'n' natural number using repeat loop.	5 5	2 3	1 1
3.	a) Write a program to create a matrix using cbind() and rbind() functions. b) Explain all R data structures with examples and use cases.	5 5	3 2	1 1
4.	a) Write function named as Factorial() using R language to find the factorial of a given number. b) Write a R program to print the sum of series (use iterative statements only) (Read input x and n) $S = 1 + x^2/2! + x^3/3! + \ldots\ldots + x^n/n!$	5 5	3 2	1 1

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